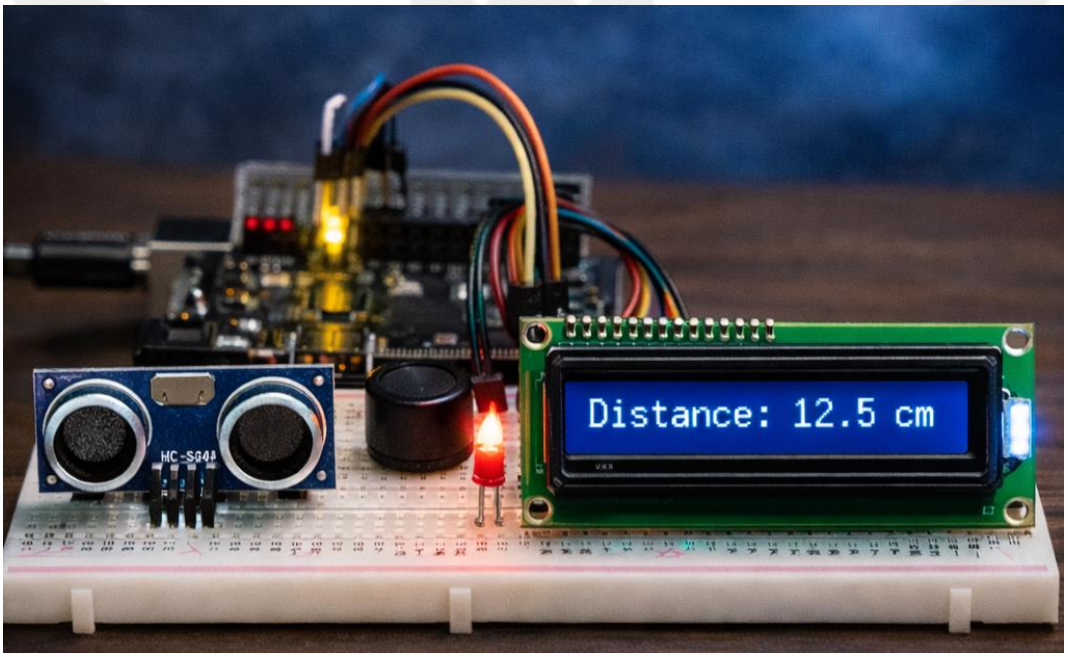
LAUNCH THE PROJECT - 02

Ultrasonic Object Detector

(Intermediate level Project)

On AYNOP® UNO Launchpad Kit



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1. Overview

This project demonstrates how to build an **Ultrasonic Object Detector** using the Arduino® UNO R4 Minima.

The system measures the distance of nearby objects using **the HC-SR04 ultrasonic sensor** and displays the measured distance on a 16×2 I2C LCD and When an object comes within a defined threshold distance, an alert is triggered.

You will learn to:

- Interface an ultrasonic distance sensor.
- Measure distance using echo timing.
- Display live data on an I2C LCD.
- Use retry logic for reliable readings.
- Activate alerts based on distance thresholds

This project introduces real-world **proximity sensing used in parking sensors, obstacle detection**, and safety systems.

2. Components Required

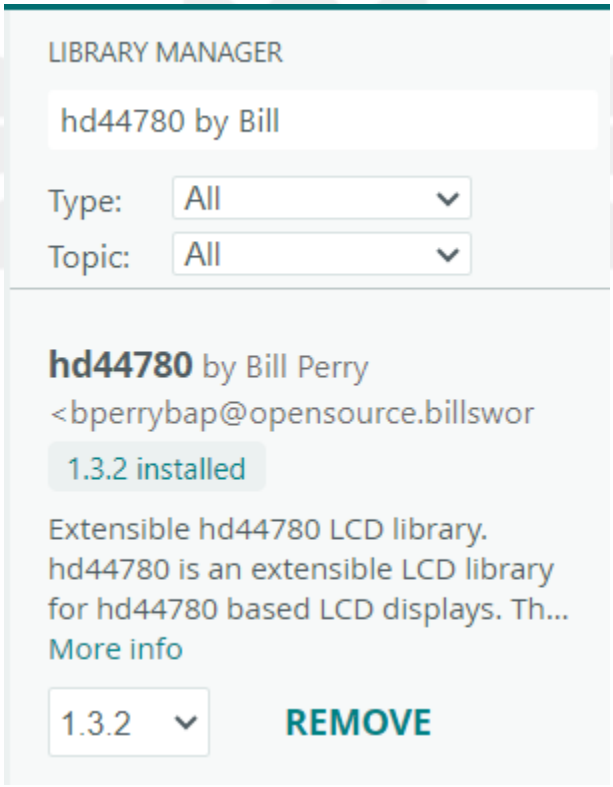
- Arduino® UNO R4 Minima board
- USB Type-C data cable
- 1 × 16x2 LCD with I2C adapter
- HC-SR04 Ultrasonic Sensor
- Active buzzer
- Jumper wires
- Breadboard

3. Software Required

- Arduino IDE (v2.3.7 or later)
- Required library – 02: **hd44780** for 16x2 I2C LCD Display

Installing the Library:

- If the **hd44780** library is already installed on your system, you may skip this step and proceed to Section 4.
- Else, Open Arduino IDE.
- Go to **Sketch → Include Library → Manage Libraries**
- Search for **"hd44780"** (by Bill Perry).
- Install the library.



Note:
We assume the **Arduino UNO R4 Minima board package** is already installed on your machine, as explained in the 00_Getting_Started/00_GettingStarted_Arduino_R4_Minima guide. If it is not installed, please refer to that document and complete the installation before proceeding.

4. Hardware Setup

This section explains how to connect the sensors and LCD to build the **Ultrasonic Object Detector.**

4.1 Wiring Diagram

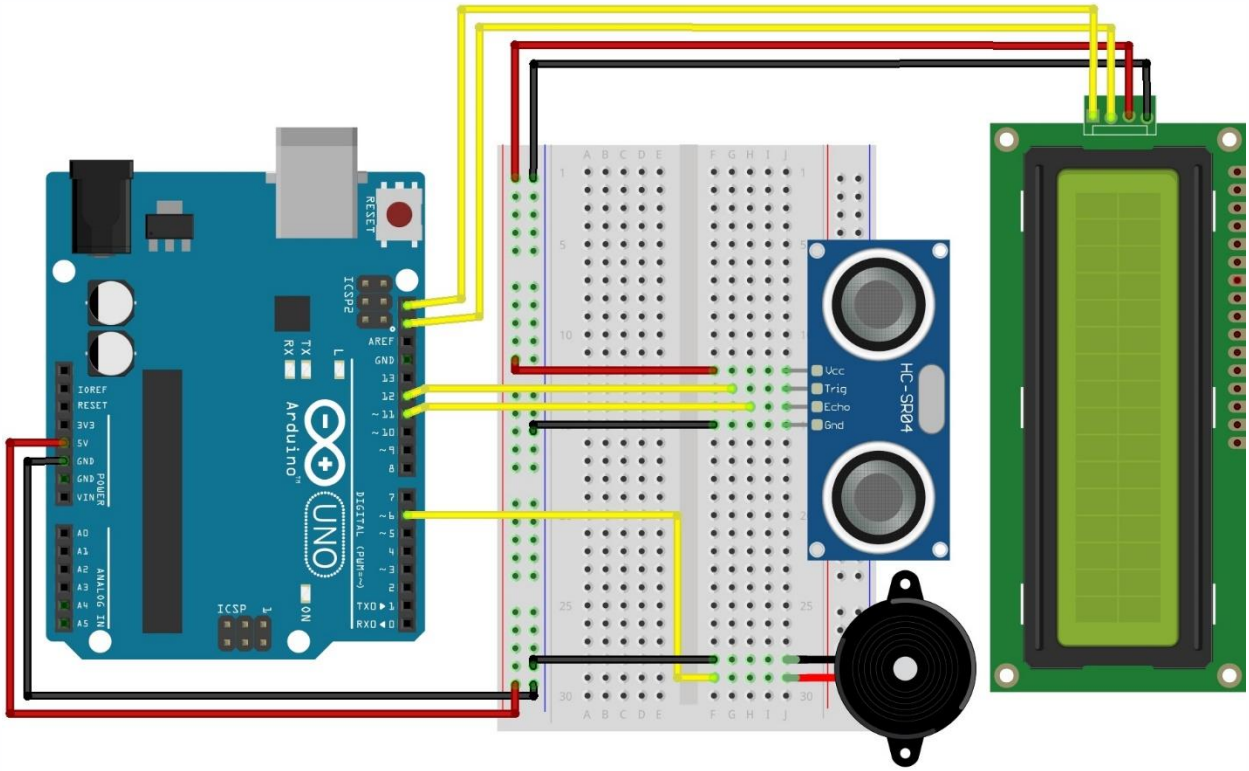


Figure 4.1 – Wiring diagram for Ultrasonic Object detector

Tip: I2C uses the same two wires (SDA, SCL) for multiple devices. Each device has a unique **I2C address**.

Tip: Always disconnect the USB cable before making or changing hardware connections.

5. Principle – How It Works

The **HC-SR04 ultrasonic sensor** measures distance using sound waves.

Distance Measurement:

- Arduino sends a short pulse to the TRIG pin.
- Sensor emits an ultrasonic signal.
- Echo returns after hitting an object.
- Arduino measures the echo time.
- Distance is calculated using:
    - Distance (cm) = (Time × 0.034) / 2

Reliability Improvement:

- To avoid false readings, up to 3 measurements are taken and the first valid distance is used.
- This improves stability and accuracy of readings/alert.

Alert Logic:

- If distance ≤ 20 cm → LED ON (On Board) + Buzzer ON.
- If distance > 20 cm → LED OFF (On Board) + Buzzer OFF

6. Procedure – Steps to Run

1. **Build the Circuit**
   a. Connect the ultrasonic sensor, LCD, and buzzer as shown in the wiring diagram (Figure 4.1).

2. **Connect the Board**
   a. Use a USB Type-C data cable to connect your UNO R4 Minima to your computer.

3. **Open the Project Code**
   a. Simply **double-click** the file *02_Ultrasonic_Object_Detector.ino* in the project folder, and it will open directly in the Arduino IDE (if installed).

4. **Confirm Board Selection**
   a. The IDE usually auto-detects the UNO R4 Minima if the package is installed.
   b. If not installed, refer to the *00_Getting_Started/00_GettingStarted_Arduino_R4_Minima* document to install the necessary board package.
   c. Verify that *Arduino UNO R4 Minima* is selected in the IDE's board selector (top toolbar).

5. **Upload the Code**
   a. Click the **Upload** button (arrow icon) in the top-left corner of the IDE.
   b. Wait until the console displays "**Done uploading**."

6. **Observe the Behaviour**
   a. The LCD will begin displaying the measured distance in centimetres.
   b. If an object is detected within approximately 90 cm, the distance value is shown.
   c. If no object is detected within approximately 90 cm range, the LCD displays "No Object Detected".
   d. If an object is detected under approximately 20 cm, the distance value is shown on LCD and it will provide an alert using buzzer and on-board LED.

7. Expected Output

- **Object Far from Sensor (above ~90–100 cm):**
  - Distance value may not be detected reliably.
  - LCD displays **"No Object Detected"**.
  - LED OFF.
  - Buzzer OFF.

- **Object at Medium Distance (~50 cm):**
  - Keep an object (for example, your hand) approximately **50 cm** away from the ultrasonic sensor.
  - LCD displays the actual measured distance of the object.
  - LED OFF.
  - Buzzer OFF.

- **Object Very Close (below 20 cm):**
  - Move the object (for example, your hand) closer than **20 cm** to the ultrasonic sensor.
  - LCD displays the measured distance and the system triggers the alert.
  - LED turns ON.
  - Buzzer sounds continuously.

💡 **Tip:** Slowly move your hand or an object toward the sensor and observe the distance value updating in real time on the LCD.

8. Code

The source code for this project is included in the downloaded folder:

📂 uno-launchpad-kit/02_Intermediate_Projects/ 02_Ultrasonic_Object_Detector/ 02_Ultrasonic_Object_Detector.ino

👉 **Tip:**

- To open the project, simply **double-click the .ino file**. If the Arduino IDE is installed, it will launch automatically and load the code.
- If you **haven't installed the Arduino IDE yet**, please refer to: 📂 uno-launchpad-kit/00_Getting_Started/00_GettingStarted_Arduino_R4_Minima to **download and install it.**

8.1 Function References

- setup() – Runs once when the board is powered on or reset. Used to initialize sensors, LCD, and I/O pins.
- loop() – Runs continuously after setup () finishes. Reads sensor values and updates the LCD at fixed intervals.
- lcd.begin(cols, rows) – Initializes the 16×2 I2C LCD.
- pinMode(pin, mode) – Configures a pin as INPUT or OUTPUT.
- digitalWrite(pin, value) – Turns the LED and buzzer ON or OFF
- lcd.backlight() – Turns ON the LCD backlight for better visibility.
- lcd.clear() – Clears all text currently displayed on the LCD.
- lcd.setCursor(column, row) – Sets the cursor position on the LCD screen.
- lcd.print() – Displays text or numeric values on the LCD.
- readDistance() – Custom function that triggers the ultrasonic sensor, measures echo time, calculates distance in centimeters, and returns a valid value or –1 if no object is detected.
- pulseIn(pin, value, timeout) – Measures the time taken for the echo pulse to return from the ultrasonic sensor.
- delay(ms) – Pauses program execution for a fixed time interval between sensor updates.

📚 **Library Reference:**

📎 Wire Library — I2C communication library used internally by hd44780.
📎 hd44780 — I2C LCD control library used to interface the 16×2 display.

📚 **For more details and advanced usage, visit:**
📎 Arduino Language Reference — The official guide for all Arduino functions.

9. Troubleshooting Tips

- **Error while compiling (hd44780 library not found)?**
  - Open Arduino IDE → Library Manager.
  - Search for "hd44780 (by Bill Perry)" and install it.
  - Restart the IDE and compile again

- **LCD backlight ON but no text?**
  - Adjust the small **blue potentiometer** on the I2C adapter to set contrast.

- **LCD backlight not ON?**
  - I2C adapter boards have a small **jumper near the backlight pins**.
  - Ensure this jumper is connected; otherwise, the backlight will stay OFF

- **LCD shows nothing?**
  - Check wiring: VCC → 5V, GND → GND, SDA → D18, SCL → D19.
  - Verify the **I2C address** (0x27 or 0x3F)

- **Upload error in Arduino IDE?**
  - Verify that the correct board (**Arduino UNO R4 Minima**) is selected in the IDE.
  - Check that the correct **COM port** is chosen.

- **Buzzer not sounding?**
  - Check buzzer polarity and connection to D6.

- **Sensor always shows "No Object Detected"?**
  - Check TRIG → D12 and ECHO → D11 wiring carefully.
  - Ensure the sensor is powered with 5V.
  - Verify that the object is within the 1–90 cm detection range

- **Distance readings unstable or jumping randomly?**
  - Ensure the ultrasonic sensor faces an open area without obstacles.
  - Avoid placing your hand too close to the sensor (< 2 cm).
  - Keep wires short and connections tight to reduce noise

- **Still not working?**
  - Recheck all wiring connections carefully.
  - Disconnect power and rebuild the circuit step-by-step.
  - Test each module (LCD, Buzzer, HCSR04 Ultrasonic Sensor) individually.

10. License

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11. Support & Feedback

We value your feedback and are happy to assist with any questions, troubleshooting, or suggestions you may have.

Email: *support@aynop.com*

When sending an email, please include your kit name (AYNOP® UNO Launchpad Kit) and, if applicable, the project name in the subject line. This will help our team respond faster and more accurately. We aim to respond to all queries within 2–3 business days. Your feedback helps us improve our products and create even better learning experiences for all Arduino beginners.